On a connected H-space the two translations by any point are equivalences

The Univalent Foundations Program

hott-book:homotopy.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted lemma carries no \label in the pinned source; its identity is fixed as file plus line range plus original number, and the number 8.5.5 was obtained by counting the theorem-class environments of section 8.5 of the same file, cross-checked against two labelled neighbours of that section (cor:pis2-hopf is 8.5.2 and lem:fibration-over-pushout is 8.5.3, both matching the count).

Editorial note. The enumerated identity is used verbatim in the claim_key and must not be changed now that the material is registered.

Editorial note. The retained counter values (chapter 8, section 5, theorem offset 4) make the counted lemma print as Lemma 8.5.5, matching the counted original number.

Editorial note. The single external reference is the result that the universe of propositions is a set, whose number is taken from the label-number dump of the pinned book snapshot; it is a navigation pointer only.

Editorial scope. One original result of the HoTT Book, counted once: translations on a connected H-space are equivalences (Lemma 8.5.5).

Allowed standard prerequisites (editorial).

- H-space. An H-space is a pointed type with a multiplication mu, a unit e and the unit laws mu(e,a) = a and mu(a,e) = a.. The unit laws are used at source line 1348 to identify mu(e,-) with the identity.
- Connected type. A is connected when trunc 0 A is contractible.. It is what makes the family over trunc 0 A trivial at the basepoint.
- Truncation of a proposition-valued family. Since prop is a set, a family P : A -> prop descends to a family P' : trunc 0 A -> prop with P'(|a|_0) := P(a).. It is the descent performed at source lines 1343-1345.

Source reference. Original theorem 7.1.11: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1793–1793. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Lemma 8.5.5. *Let A be a connected H-space. Then for every $a : A$, the maps $\mu(a, -) : A \to A$ and $\mu(-, a) : A \to A$ are equivalences.*

Proof. Let us prove that for every $a : A$ the map $\mu(a, -)$ is an equivalence. The other statement is symmetric. The statement that $\mu(a, -)$ is an equivalence corresponds to a type family $P : A \to \mathsf{Prop}$ and proving it corresponds to finding a section of this type family.

The type Prop is a set (original theorem 7.1.11) hence we can define a new type family $P' : \|A\|_0 \to \mathsf{Prop}$ by $P'(|a|_0) :\equiv P(a)$. But A is connected by assumption, hence $\|A\|_0$ is contractible. This implies that in order to find a section of P', it is enough to find a point in the fiber of P' over $|e|_0$. But we have $P'(|e|_0) = P(e)$ which is inhabited because $\mu(e, -)$ is equal to the identity map by definition of an H-space, hence is an equivalence.

We have proved that for every $x : \|A\|_0$ the proposition $P'(x)$ is true, hence in particular for every $a : A$ the proposition $P(a)$ is true because $P(a)$ is $P'(|a|_0)$. □